Fiscal Year:	FY 2007	**Task Last Updated:**	FY 05/21/2007
PI Name:	Niesel, David William Ph.D.		
Project Title:	Streptococcus pneumoniae Gene Expression and Virulence Potential in the Space Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77555-1019	**Congressional District:**	14
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	02/01/2000	**End Date:**	01/31/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-118/ISS NOTE: Received extension to 1/31/2010 (previously 1/31/2009) per A. Chu (2/2009) NOTE: Received extension to 1/31/2009 per A. Chu/ARC (6/2008) 6/23/06--End date is 7/31/2008, per A. Chu (ARC) renewed. Updated Aug. 26, 2005.		
Key Personnel Changes/Previous PI:	Nina Williams - Research assistant / Graduate Student		
COI Name (Institution):	Watson, David (The University of Texas Medical Branch)		
Grant/Contract No.:	NCC2-1160		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Extended space flight and the long-term habitation of space environments are the new focus of the U.S. space effort. While we have learned much about the space environment, extended residence in space represents a challenge to maintain crew members' health. Evidence has accumulated about alterations in the immune system in the space environment and little is known as to the effect of this unique environment on the physiology, gene expression or the virulence potential of microbes that cause infectious disease. In our study, we are investigating the human opportunistic pathogen, Streptococcus pneumoniae. S. pneumoniae is an important human pathogen and the leading cause of bacterial pneumonia, meningitis, and otitis media. Importantly, this pathogen has been isolated previously from the crew pre-flight and related bacteria are found in the spacecraft environment. During long term missions, this opportunistic pathogen represents a concern for crew members. Specifically, we will determine unique gene and protein expression and assess the virulence potential of this opportunistic pathogen under conditions of low shear modeled microgravity (LSMMG) and in the space environment. The experimentation has been organized to address three specific objectives. Experiments will be performed to identify and characterize S. pneumoniae genes and proteins which are differentially expressed in response to the space environment or in response to LSMMG. The second objective will be to investigate the expression of known S. pneumoniae virulence genes or properties in LSMMG. Finally, we will determine if S. pneumoniae genes differentially expressed in LSMMG or during space flight are also expressed during animal infection. This experimental approach will result in new information about a significant human pathogen, add to our knowledge the S. pneumoniae pathogenic mechanism and also provide basic information on bacterial model system of gene and protein expression in the space environment. These studies will also provide for a rationale risk assessment for this organism and other opportunistic microbes in the context of long-term space flight and habitation and contribute to our understanding of the virulence of this bacterium on earth. Thus, this experimentation will benefit NASA objectives, public health agencies and public and private commercial concerns.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study will reveal new information about S. pneumoniae, a significant opportunistic bacterial pathogen. The data collected will advance our understanding of the pathogenic mechanism of this bacterium and how it adapts to different environments. Understanding how this microbe responds to the space environment will additionally allow a rationale risk assessment for this and other opportunistic bacterial pathogens in the space environment. This could be important in the planning and execution of long-term space missions and for the habitation of space. The space flight results will benefit NASA's goals of long-term human space travel and potentially provide information that could lead to development of countermeasures that crew members could use on-orbit to detect illnesses or treatment diseases on-orbit. Additionally, new information from these studies on the S. pneumoniae pathogenic mechanism will have direct earth benefits in terms of understanding the diseases mediated by this pathogen and could lead to novel vaccines and therapeutics for this bacterial pathogen.
Task Progress:	This past year, we have devoted a significant amount of time in preparing for flight. We are currently preparing for the 13A.1 and STS 118 flight opportunity. We continue to be productive with both flight preparations and ground experimentation and have modified our approaches/experimental design to take advantage of this opportunity. For the SPEGIS experiment, new hardware was developed and extensively tested this past year. Stowage and storage conditions and equipment requirements have been evaluated to take advantage of a shuttle mid-deck and ISS experimental opportunity. Our experimental approach was to optimize cultures of Streptococcus pneumoniae 6304 to remain viable for extended periods in broth at 40C. Hardware was developed at NASA Ames which utilizes commercial culture tubes (Sarstadt, Inc.) The coupling of this hardware and the culture tubes with an O-ring seal provides for required three levels of containment of the bacterial culture. Extensive testing has defined optimal times and limits for stowage conditions for flight. Viability and stowage requirements for S. pneumoniae during launch phase, cold stowage prior to the initiation of growth, growth parameters to provide sufficient bacterial numbers for analysis, and the optimal storage for sample return to earth for laboratory analyses required extensive experimentation and were generally defined. Overall, the experimental design utilizes three canisters, each which contains culture tubes with different bacterial concentrations. A single incubation time of 15 hours 50 minutes has been established and the use of multiple concentrations will allow recovery of mid-logarithmic and late logarithmic phase cells. Since bacterial cells will be frozen after growth, we will return to the originally designed experimentation which utilizes live bacterial cells. This will allow us to evaluate mouse virulence as well as focusing on global transcriptional profiling using DNA microarrays and characterization of the proteome in the space environment. We have worked closely with engineering at NASA Ames, the Cold Stowage group at JSC, and Dan Connor and the MERLIN group at University of Alabama – Birmingham. We have incubated S. pneumoniae 6304 in the MERLIN two different times at UAB in the past year. This allowed an optimal rehearsal and refinement of flight procedures in preparation for EVT and FTR. Results from this testing has lead to the development of aluminum tube sleeves, refinement in the machining of the canisters, modification of the foam holders and optimization of procedures for culture handling both pre-and post-incubation. We now understand much more about the cooling and heating of the samples and the hardware. We completed an EVT in January using facilities at NASA JSC and our laboratory at UTMB. The EVT was a success in terms of validating our methods and approach and defining conditions for optimal production of cells at the desired concentration in flight. Data was collected and we are in the process of analyzing this data which will be valuable as a baseline for flight and ground-based control samples. In March, we conducted a FTR at NASA KSC under conditions that we will use for flight. Experimentation was conducted at the Space Life Sciences laboratories. The SPEGIS team from Ames and UTMB was present for the FTR. The laboratory was "set up" and the team became familiar with the facility and the lab. The FTR allowed further refinement of the ATOP document and allowed practice of the entire protocol leading up to turnover and launch. Most importantly, the FTR allowed refinement of the procedures and practice of SPEGIS members functioning as a team. In ground-based experiments, we continue to expand gene and protein experimentation using low shear modeled microgravity (LSMMG), produced with high aspect rotating vessels (HARVs). Experiments have shown protein and gene expression differences between S. pneumoniae cultured under LSMMG and laboratory control conditions. One manuscript has been published showing alterations in global transcriptional activity under modeled microgravity and the second will describe alterations in the virulence activity under LSMMG. We are also assessing the total proteome from LSMMG and lab control cultures. Previous results have shown 12 and19 protein spots to be under expressed and over

	expressed respectively after LSMMG culture. MALDI-TOF sequencing of two over expressed protein spots which are in the same transcriptional unit as dihydrofolate reductase (dfr) and the pneumococcal surface antigen A (PsaA). Significantly, Dfr contributes to trimethoprim resistance in S. pneumoniae and PsaA is a major adhesin. Our experiments will document changes in global transcription and protein expression in land based models, which will be used to compare protein and gene expression between LSMMG, space flight and lab cultures. Changes in gene and protein expression profiles under flight conditions may indicate the potential of the bacterium to initiate infection or produce disease in these environments. We believe that this would have significant implications for long-term space flight or space habitation.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
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